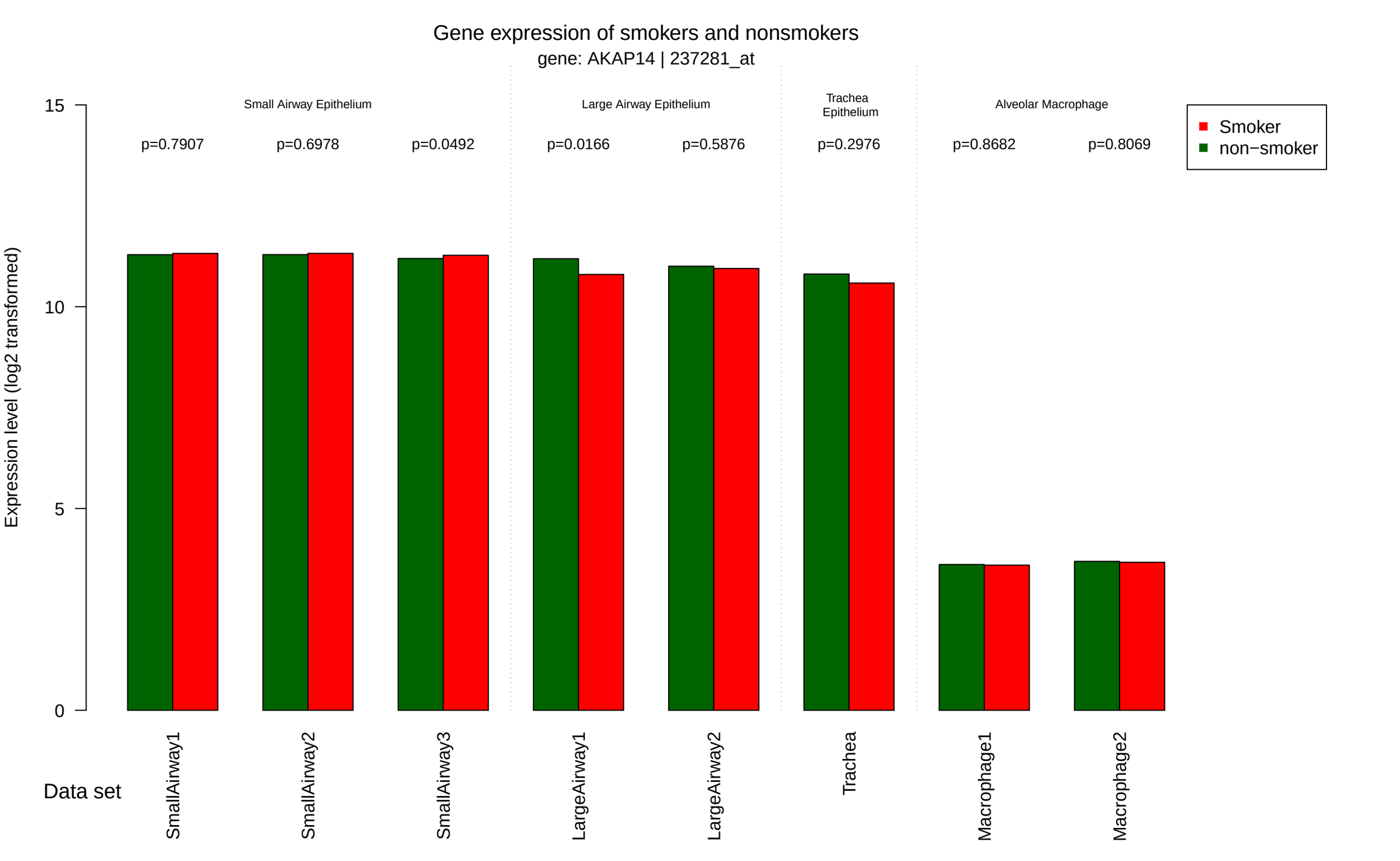
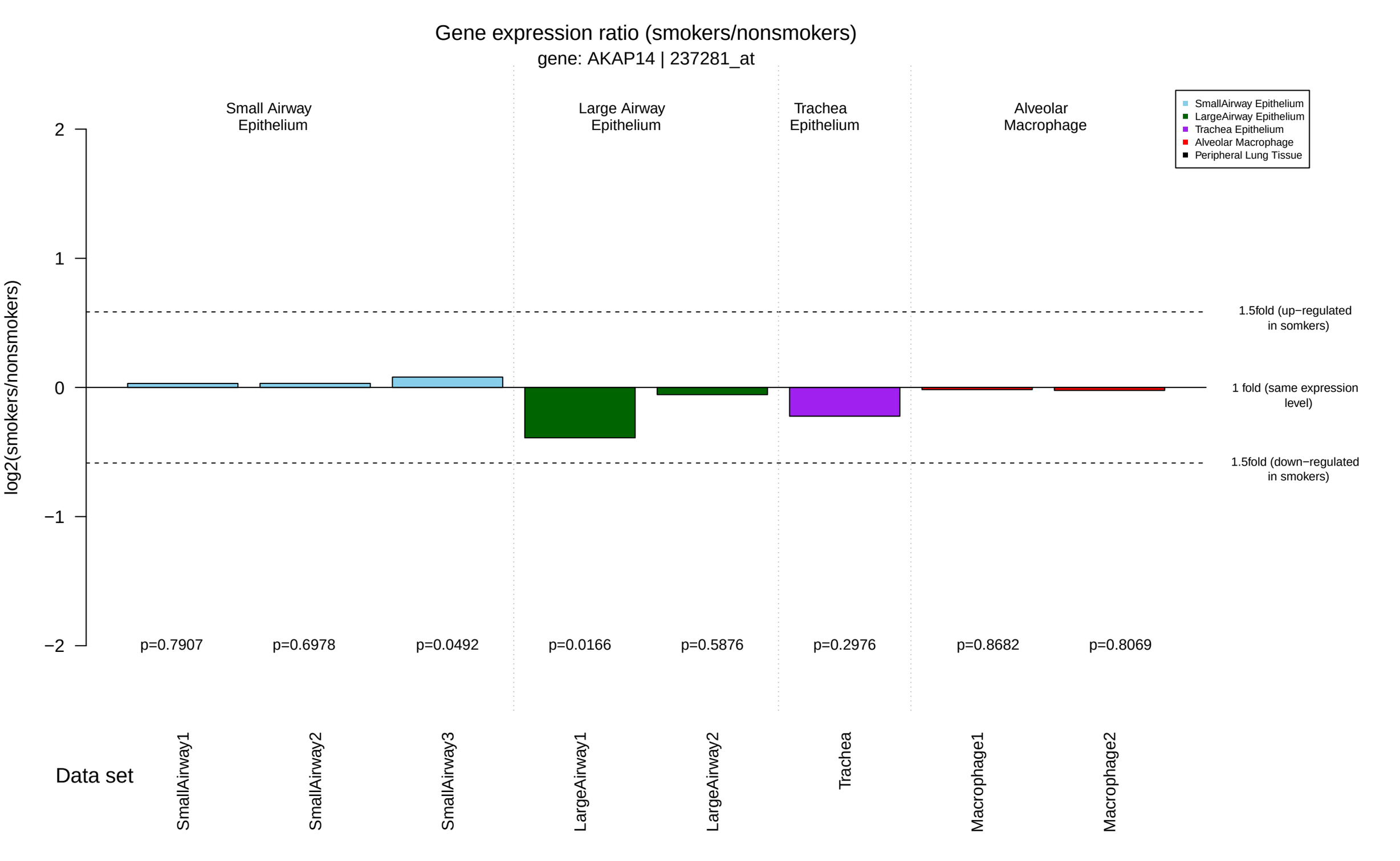
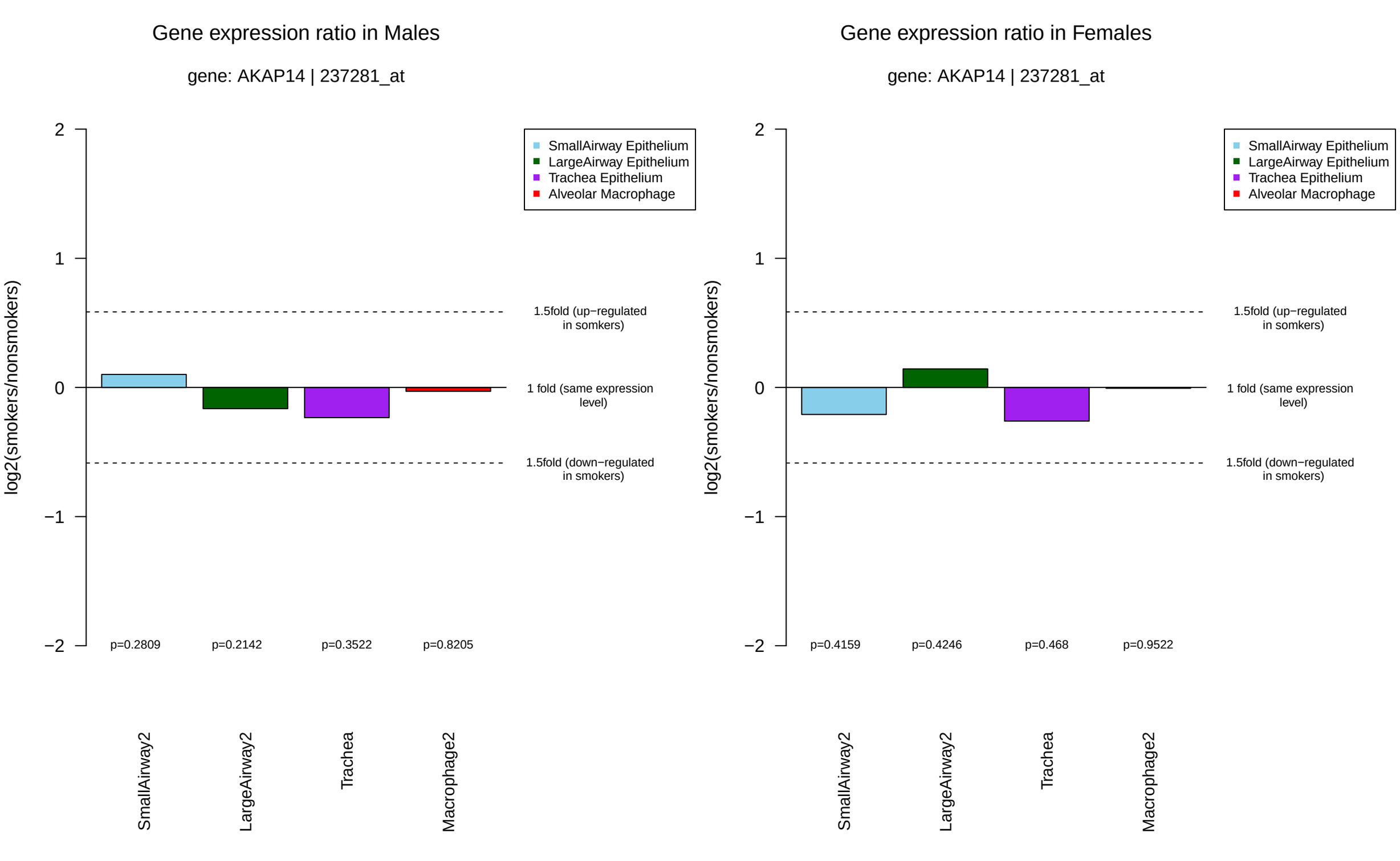
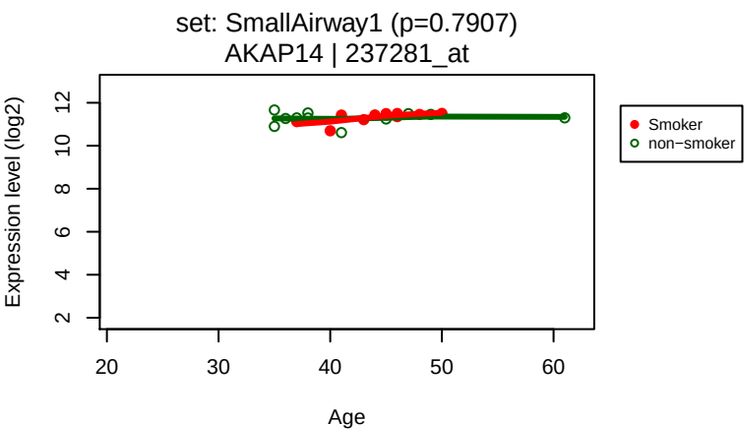
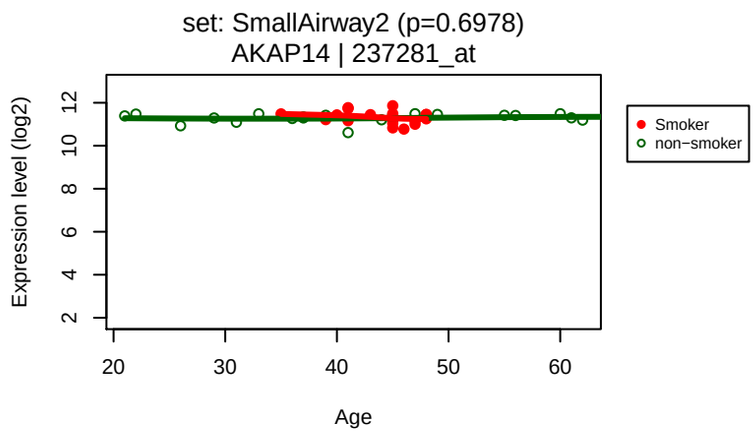
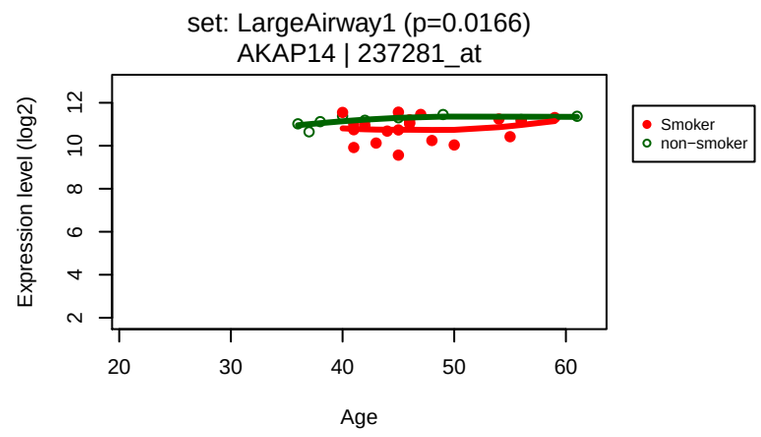

Gene expression of smokers and nonsmokers
gene: AKAP14 | 237281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7907
p=0.6978
p=0.0492
p=0.0166
p=0.5876
p=0.2976
p=0.8682
p=0.8069
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

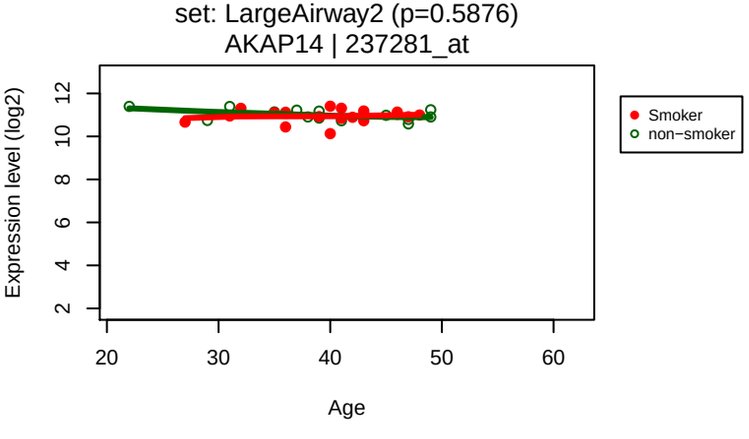

Gene expression ratio (smokers/nonsmokers)
gene: AKAP14 | 237281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7907
p=0.6978
p=0.0492
p=0.0166
p=0.5876
p=0.2976
p=0.8682
p=0.8069
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

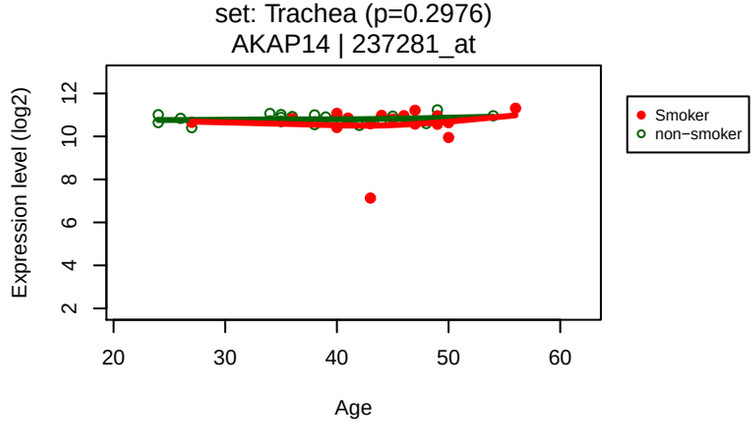

Gene expression ratio in Males
gene: AKAP14 | 237281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2809
p=0.2142
p=0.3522
p=0.8205
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP14 | 237281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4159
p=0.4246
p=0.468
p=0.9522
SmallAirway2
LargeAirway2
Trachea
Macrophage2

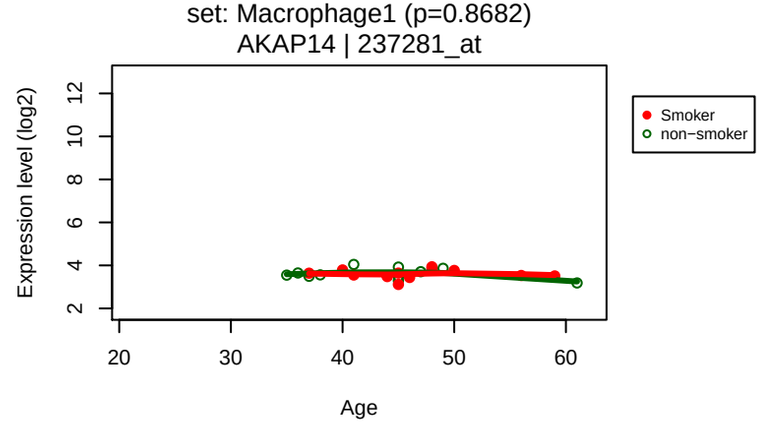

set: SmallAirway1 (p=0.7907)
AKAP14 | 237281_at
Expression level (log2)
Age
Smoker
non−smoker

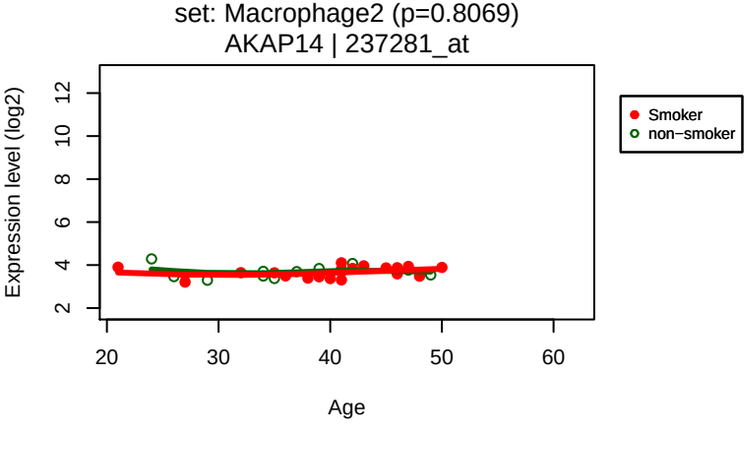

set: SmallAirway2 (p=0.6978)
AKAP14 | 237281_at
Expression level (log2)
Age
Smoker
non−smoker

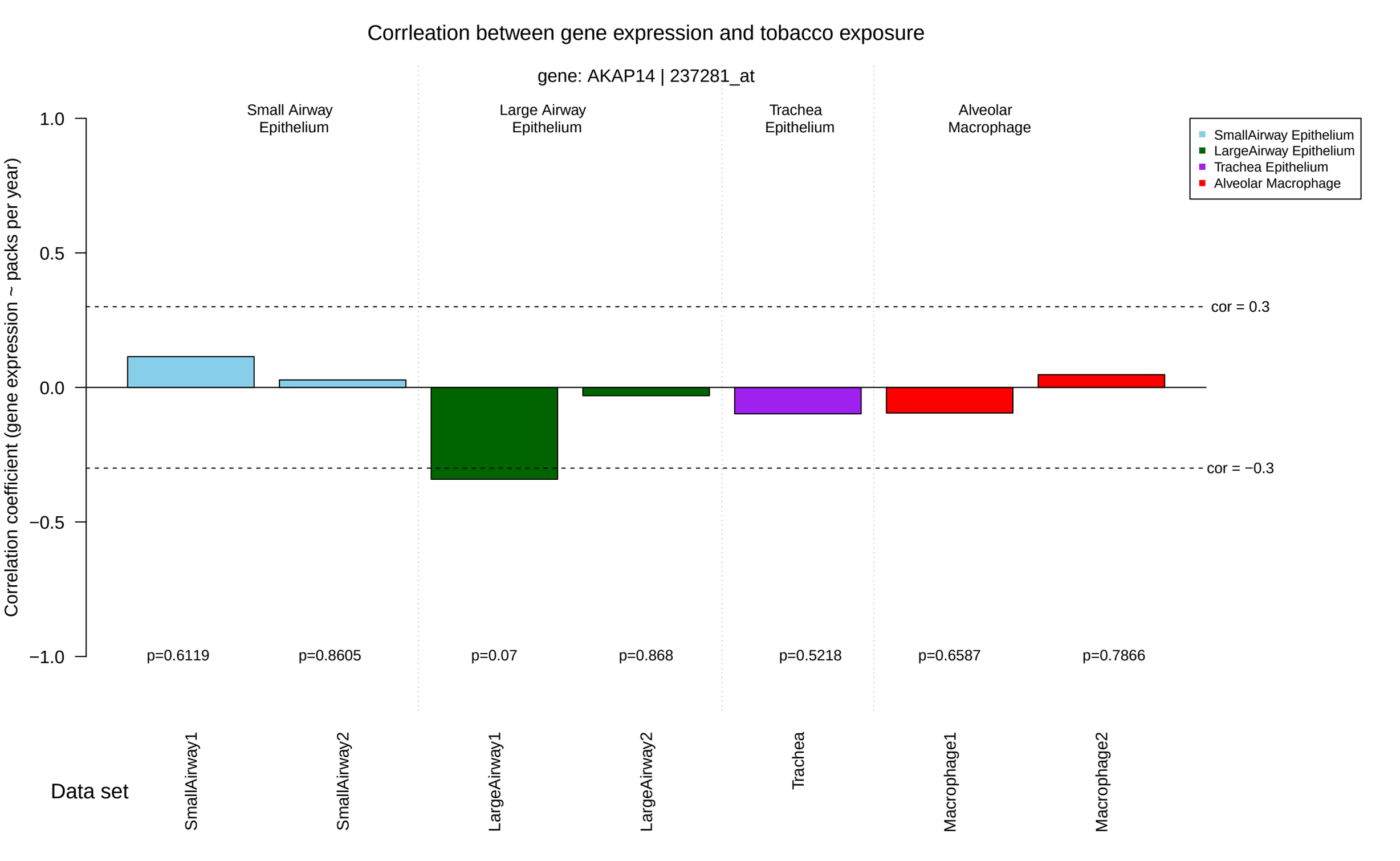

set: LargeAirway1 (p=0.0166)
AKAP14 | 237281_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5876)
AKAP14 | 237281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2976)
AKAP14 | 237281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8682)
AKAP14 | 237281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8069)
AKAP14 | 237281_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP14 | 237281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6119
p=0.8605
p=0.07
p=0.868
p=0.5218
p=0.6587
p=0.7866
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2